Vrishank N Amembal

B.Tech Computer Science & Engineering | PES University (2023-2027)
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Profile

Computer Science and Engineering undergraduate with hands-on project experience in software development, machine learning, and system integration. Experienced in building technical solutions involving data processing, predictive modelling, and interactive visualization. Interested in developing reliable, maintainable software and strengthening engineering skills through collaborative product development.

Experience

Summer Intern — PESU IoT Centre (Jun 2026 – Aug 2026)

- Worked on ECG preprocessing, ML-based cardiac event detection, and P/QRS/T wave segmentation.
- Integrated ECG-derived events with virtual and physical cardiac simulation with a visual dashboard representation.

Event Management Team Member — Embrione Club (Sep 2024 – Oct 2024)

- Planned and coordinated club events, handling logistics and on-ground execution as part of a team.

Projects

ECG-Based Cardiac Event Detection & Simulation (Undergraduate capstone)

- Built an end-to-end system that takes raw ECG through signal processing and machine learning to detect cardiac events.
- Connected the detected events to a Unity-based virtual heart model for cardiac simulation.

Short-Term Energy Load Forecasting · IndiaAI Impact Gen-AI Hackathon, 14th among 3,300+ participants

- Developed a short-term electricity load forecasting solution using time-series forecasting and machine learning.
- Explored Time-Series Foundation Models (TSFMs) and LightGBM for predictive energy demand forecasting and model ensembling.

3D Campus Sports Complex Visualization

- Recreated the PES University Sports Complex in a virtual 3D space, covering modelling, scene design, and visualization.

Skills

- **Programming Languages:** C++, Python, C, JavaScript
- **Web Technologies:** HTML, CSS, React, Typescript, Nodejs
- **Machine Learning & Data:** NumPy, scikit-learn, PyTorch, LightGBM
- **Development & Visualization:** Unity, Blender, OpenGL
- **Tools & Version Control:** Git, GitHub

Publications

Physiologically Guided R-Centered Deep Learning for ECG Wave Delineation across Heterogeneous Databases.

- Accepted at the 13th International Conference on Biomedical and Bioinformatics Engineering, Hokkaido.

Leadership & Certifications

- **Vice President Public Relations**, Toastmasters Club: managed promotional posters and social-media content.
- **Nautanki Dramatics Team**: represented PES University at IIT Kharagpur
- **Certifications**: Artificial Intelligence and Machine Learning (Pravega, IISc); Hands-on Arduino: Real-World Applications (PESU I/O).